



Risk-Adaptive Neurosymbolic Intent Planning for Optical Networks

A Pre-Deployment Decision Mechanism with Joint Semantic and QoT Assessment

Candidate: Felipe Abadía

Academic Advisor: Prof. Massimo Tornatore

Dipartimento di Elettronica, Informazione e Bioingegneria — Politecnico di Milano

OUTLINE

[01] Bottleneck

The Optical Intent Planning Bottleneck

Physical-layer constraints, token saturation, and hallucinated routing in optical backbones

[02] Architecture

Neurosymbolic Intent Planning Pipeline

Decoupling probabilistic reasoning (NL to PDDL) from deterministic solvers and physics tools

[03] Risk Gates

Pre-Deployment Risk Gates: Semantic & Physical Validation

Sequential fail-fast decision via Layer 1/2 semantic gate (U_sem) and GN-model QoT gate

[04] Evaluation

Experimental Testbed Validation on 17-Node Optical Topology

Benchmarking safety, human intervention reduction, and orchestration latency against baselines

[05] Outlook

Key Takeaways, System Guarantees & Future Directions

Summary of thesis contributions, operational guarantees, and extension to joint compute scheduling

Motivation: The Vision of Intent-Based Optical Networks

⚠ Operational Shift: Manual Bottleneck

- Optical backbones carry terabits of core traffic across ROADM networks
- Traditional workflow: manual CLI scripts and complex RESTConf payloads
- Human configuration delays lightpath provisioning by hours or days
- High cognitive load and misconfiguration risk across multi-vendor links

🎯 Goal: Transition to autonomous Intent-Based Networking (IBN)

⚡ Operational Promise: Autonomous Vision

- High-level abstraction: specify WHAT is needed, not HOW to configure it
- Carrier intent: 'Establish a 400G lightpath between Milan and Rome avoiding link L2'
- Autonomous translation into verified, collision-free physical lightpaths
- Rapid sub-second provisioning reducing operational delays by orders of magnitude

⚠ Critical Challenge: Optical networks do not tolerate probabilistic errors

Operational Flow: [Operator NL Intent] → [AI Intent Orchestrator] → [Zero-Error Physical Lightpath]

Illustrative Failure: Why Standard LLMs Break Optical Backbones



1. Token Budget Saturation

Massive topology dumps trigger attention degradation, dropping critical operator route exclusions



2. Hallucinated Physics

Probabilistic predictors lack wave propagation engines, violating non-linear GSNR and noise margins



3. Semantic Drift

Unconstrained conversational refinement lacks convergence bounds, mutating initial boundary constraints



4. Reactive Latency

Trial-and-error configuration risks live service disruption and introduces high control-plane recovery latency



5. Suboptimal HITL

Binary all-or-nothing review causes operator fatigue or outages; models fail to fail-early on ambiguity

Empirical Risk: Unconstrained LLMs allow up to 34% unfeasible deployments, semantic drift loops, and critical control-plane latency

Problem Statement: Given, Decide, Objective & Constraints



1. Given (System Inputs)

- Unstructured operator intent: $\mathcal{I}_{NL}$
- Active optical network graph: $\mathcal{G}(V, E)$ via RESTConf
- Physical parameters: span length L , attenuation α , gain G_{amp}
- Feasibility target: $GSNR_{th} = SNR_{min} + \text{Margin}$



2. Decide (Variables & Actions)

- Formal symbolic specification: $\mathcal{S}_{PDDL}$ from intent
- Optimal physical lightpath route: $\pi \in \mathcal{K}_{path}$ (K -SP)
- Pre-deployment control action: $a \in \{\text{approve, clarify, replan}\}$
- Provisioning routing configuration: c dispatched to controller



3. Objective (Optimization Goal)

- Minimize composite operational friction: $\min \mathcal{J} = \alpha \cdot N_{hitl} + \beta \cdot T_{tokens}$
- Cut operator fatigue: $\min N_{hitl}$ (minimize interruptions)
- Bound compute cost and latency: $\min T_{tokens}$ (minimize prompt tokens)
- Hard safety guarantee: Pre-deployment $\mathcal{D}(U_{sem}, QoT_{valid}) = \text{approve}$



4. Constraints (Resource Limits vs. Physical & Semantic Boundaries)



Resource Constraints (System & Solver Limits)

- Localized prompt context window bound: $T_{prompt} \leq T_{max} \ll T_{full}$ (via G_{sub})
- Strict end-to-end execution latency budget: $t_{exec} \leq t_{max_budget}$
- Symbolic solver path search complexity: bounded K -SP with $K \in [3, 5]$



Boundary Constraints (Physical & Semantic Feasibility)

- Zero semantic drift tolerance bound: $U_{sem} \leq \tau_{sem}$
- Deterministic optical QoT feasibility: $GSNR(\pi) \geq GSNR_{th} \wedge P_{rx}(\pi) \geq P_{rx,min}$
- Hard pre-deployment physical feasibility state: $QoT_{valid} \in \{0, 1\}$

Proposed Solution: Neurosymbolic Decoupling

Neural Subsystem (Semantic Domain)

Linguistic Compilation & Early Semantic Gate (Sections 3.2.1 & 3.3.1)

- **Linguistic Compiler:** Translates operator intent to formal PDDL ($\mathcal{I}_{NL} \rightarrow \mathcal{S}_{PDDL}$)
- **Strict Separation:** Prohibited from arithmetic, graph routing, or physics calculation
- **Early Fail-Fast Gate:** Audits semantic uncertainty ($U_{sem} \leq \tau_{sem}$) before physics simulation
- **Hallucination Suppression:** CFG grammar validator (v_{struct}) eliminates syntax errors

Symbolic Subsystem (Optical Domain)

Deterministic Path Search & QoT Validation (Sections 3.2.1 & 3.3.1)

- **Deterministic Routing:** Yen's K -SP explores topologically valid paths over G_{sub}
- **Analytical Physics Engine:** Coherent GN model computes non-linear GSNR and P_{rx}
- **Late Physical Risk Gate:** Validates transmission feasibility ($QoT_{valid} = 1$)
- **Zero Unfeasible Deployments:** Guarantees 100% physical safety before provisioning

Core Thesis Contributions (Architectural Novelties)

1. Strict Neurosymbolic Compilation

Formal Intent Translation & CFG AST Validator

- Translates NL intent into formal PDDL ($\mathcal{I}_{NL} \rightarrow \mathcal{S}_{PDDL}$)
- Deterministic CFG AST validation blocks structural hallucinations ($v_{struct} \in \{0, 1\}$)

2. Scoped Optical GraphRAG

Topological Context Scoping via RESTConf

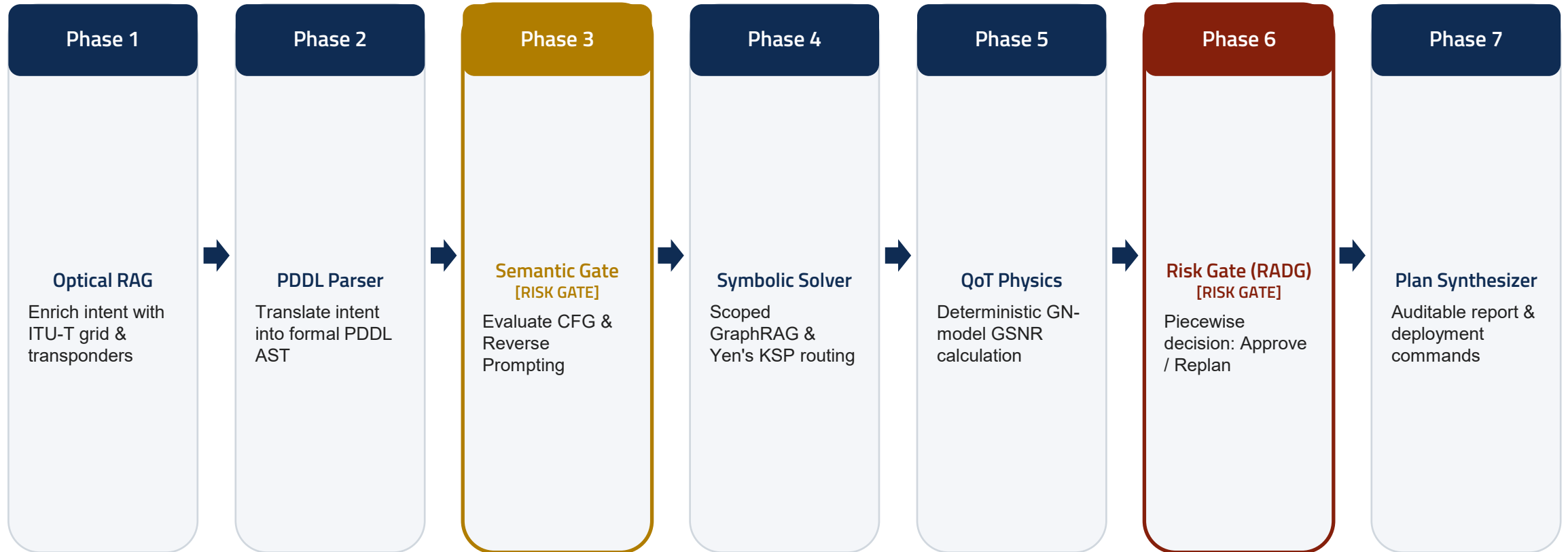
- Dynamically extracts localized ellipsoid subtopologies ($G_{sub} \subseteq G$)
- Eliminates token saturation with $> 75\%$ prompt token reduction

3. Fail-Fast Risk Gates (RADG)

Pre-Deployment Orthogonal Decision Function

- Orthogonal sequential validation: Gate 1 ($U_{sem} \leq \tau_{sem}$) & Gate 2 ($QoT_{valid} = 1$)
- Guarantees 100% optical safety and cuts operator HITL fatigue by $> 70\%$

End-to-End System Architecture & Pipeline Flow



⏸ Phase 3b: Clarify Loop (LangGraph interrupt())

If $U_{sem} > \tau_{sem}$: Pauses execution and prompts operator for intent disambiguation

🔄 Phase 6: Suggest Replan Loop (Physics Unfeasible)

If $QoT_{valid} = 0$: Suggests operator to relax constraints (lower baud rate, alternate link)

Overcoming Token Saturation: Scoped Optical GraphRAG

Deterministic Ellipsoid Subtopology Scoping

- Full topology dumps overwhelm LLM context windows
- Raw JSON contains excessive telemetry: ROADM ports, EDFAs, fibers
- Mock GraphRAG extracts only the dynamic ellipsoid neighborhood between endpoints
- Filters out irrelevant core subnets, links, and unused transponders
- Quantitative Impact: Over 75% reduction in prompt token payload
- Sub-millisecond graph extraction: $O(V + E)$ executed in pure Python
- Guarantees sharp LLM attention focus on active optical constraints



Visual Diagram: Full 17-Node Backbone vs Scoped Ellipsoid Subtopology

(Target: 4:3 Network Topology & Subtopology Diagram — Replace in PowerPoint)

Overcoming Semantic Drift: Reverse Prompting & HITL

Two-Layer Semantic Uncertainty (U_{sem})

- Layer 1 (Structural): CFG AST validator instantly catches syntax errors
- Layer 2 (Semantic): Reverse Prompting reconstructs NL directly from PDDL
- Independent LLM judge measures semantic discrepancy d_{sem} in $[0, 1]$

Uncertainty Formulation:

$$U_{sem} = \begin{cases} 1 & \text{if } v_{struct} = 0 \\ d_{sem} & \text{if } v_{struct} = 1 \end{cases}$$

Fail-Fast HITL Clarification Loop

- Evaluated BEFORE invoking graph solvers or physical tools
- If $U_{sem} > \tau_{sem}$: Pipeline pauses via LangGraph interrupt()
- Prompts operator to clarify ambiguous parameters or missing nodes
- Eliminates infinite trial-and-error conversational loops
- Guarantees that downstream tools receive mathematically verified intent
- If operator approves valid syntax: Bypasses parsing straight to solver

Fail-Fast Guarantee: Zero computational waste on physics simulation when intent is ambiguous

Pre-Deployment Risk Gate: The RADG Decision Function

Piecewise Decision Formulation $D(U_{sem}, QoT_{valid})$:

$$D(U_{sem}, QoT_{valid}) = \begin{cases} \text{clarify} & \text{if } U_{sem} > \tau_{sem} \\ \text{replan} & \text{if } U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 0 \\ \text{approve} & \text{if } U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 1 \end{cases}$$

🔍 Clarify Intent

Condition: $U_{sem} > \tau_{sem}$

- Trigger: Semantic uncertainty exceeds threshold
- Action: Pause pipeline via interrupt()
- Prompts operator to provide missing data
- Prevents unverified intent from reaching physics

🔄 Suggest Replan

Condition: $U_{sem} \leq \tau_{sem} \wedge QoT = 0$

- Trigger: Valid semantics, but physics infeasible
- Action: Physics failed; notifies operator
- Suggests relaxing constraints (e.g. lower baud rate)
- Loops back to PDDL parsing with feedback

✓ Auto-Approve

Condition: $U_{sem} \leq \tau_{sem} \wedge QoT = 1$

- Trigger: Clear semantics and feasible GSNR
- Action: Autonomous zero-touch provisioning
- Generates auditable planning report
- Zero unverified configurations reach controller

Deterministic Physical Layer: GN-Model QoT Validation

Analytical Gaussian Noise (GN) Engine

- Pure Python implementation: zero LLM arithmetic

Accumulated ASE Noise Power per Span:

$$P_{ASE,m} = (G_m - 1) \cdot h\nu \cdot R_s \cdot NF_m$$

Nonlinear Interference (NLI) Noise Power:

$$P_{NLI,m} = \eta_0 \cdot L_{eff}^2 \cdot P_{ch}^3$$

- Accounts for fiber Kerr nonlinearity, dispersion, and EDFA noise

Optical Feasibility & GSNR Criteria

Generalized Signal-to-Noise Ratio (GSNR):

$$GSNR = \frac{P_{ch}}{P_{ASE} + P_{NLI}} \geq GSNR_{th}$$

Receiver Power Sensitivity Budget:

$$P_{rx} = P_{launch} - A_{total} + G_{total} \geq P_{rx,min}$$

- Deterministic feasibility: GSNR >= threshold AND Power >= sensitivity

 Execution Speed: Under 5 milliseconds per candidate route (100% deterministic reproducibility)

Experimental Setup & Testbed Environment

17-Node German Core Network Benchmark

- Realistic telecom benchmark: 17 ROADMs and 26 fiber links
- Standard Single-Mode Fiber (SMF-28): $\alpha = 0.2 \text{ dB/km}$
- Dispersion parameter $D = 16.7 \text{ ps}/(\text{nm}\cdot\text{km})$, $\gamma = 1.2 / (\text{W}\cdot\text{km})$
- Amplified spans: dual-stage EDFAs with noise figure $F = 5.5 \text{ dB}$
- Dynamic link lengths ranging from 45 km to 350 km per span
- Orchestrator: LangGraph StateGraph with memory persistence
- Telemetry: RESTConf and Mock SDON testbed client adapters



Network Topology: Nobel-Germany 17-Node 26-Link Core Backbone

(Target: 16:9 Topology Map with ROADM Nodes & Amplified Spans — Replace in PowerPoint)

Evaluation Framework & Benchmark Scenarios



Architectural Baselines

- Baseline A (LLM-Only): Direct prompt-to-configuration generation with heuristic retry
- Baseline B (Static Rule-Based): Strict regex parser with always-on human review
- Proposed (Neurosymbolic RADG): Decoupled translation + sequential risk gates
- Evaluates LLM reasoning limits vs deterministic safety



100 Test Demands (4 Classes)

- Nominal Intents [40]: Unambiguous valid routing requests with feasible physics
- Ambiguous Intents [20]: Under-specified constraints triggering semantic gate (U_{sem})
- Infeasible Intents [25]: High modulation over long unamplified reaches (QoT fails)
- Adversarial Prompts [15]: Hallucinated nodes & grammar violations



Evaluation Metrics

- Pre-Deployment Blocking Accuracy: % of invalid routes blocked before deployment
- Human Intervention Rate: % of demands requiring human operator clarification
- End-to-End Latency: Pipeline execution time from NL intent to planning report
- Token Consumption: Prompt overhead across scoped vs full topologies

Key Findings & Pre-Deployment Guarantees

100%

Pre-Deployment Safety Guarantee

Zero unfeasible lightpaths reach controller (vs 34% violation in LLM baseline)

> 70%

Reduction in Operator Fatigue

Autonomous approval when $U_{\text{sem}} \leq \tau_{\text{sem}}$ (operator engaged only on high risk)

< 15 ms

Deterministic Physics & Solver Latency

Sub-second orchestration; solver & GN model execute in sub-milliseconds



Empirical Benchmark: GSNR Distribution & Blocking Rate vs Baselines

(Target: 16:9 Matplotlib / Chapter 4 Benchmark Plot — Replace in PowerPoint)

Conclusions & Main Takeaways

Neurosymbolic Decoupling is Essential

- Probabilistic LLMs must never calculate physical impairments or route lightpaths
- Natural language reasoning cleanly bridges to formal PDDL planning
- Guarantees formal soundness without constraining operator expressiveness

Sequential Risk Gates Prevent Failure

- Early semantic evaluation (U_{sem}) eliminates drift before physics computation
- Deterministic GN-model verification (QoT_{valid}) ensures 100% physical feasibility
- Zero unverified configurations ever reach the optical controller

Selective HITL Optimizes Efficiency

- Engaging operators proportionally to risk eliminates fatigue while maintaining safety
- Over 70% reduction in human intervention compared to always-on review
- Sub-second planning latency enables rapid dynamic lightpath turn-up

Production-Ready Engineering

- Fully implemented LangGraph state machine with memory checkpoints
- Benchmarked on realistic 17-node German optical topology and RESTConf testbed
- Establishes a reproducible foundation for autonomous optical networking

Future Outlook & Acknowledgments



Joint Compute and Optical Scheduling

Extending PDDL domain to co-schedule distributed GPU cluster workloads alongside optical lightpaths



Multi-Band & Dynamic Optical Channels

Expanding analytical GN-models to incorporate C+L multi-band SRS inter-channel Raman crosstalk



Autonomous Online Self-Healing

Ingesting real-time testbed telemetry for closed-loop intent re-planning upon physical fiber degradation

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Thank You for Your Attention!

Questions & Discussion